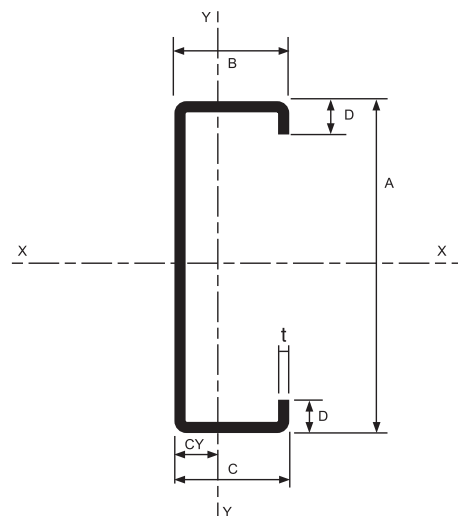


C-PURLINS

HIGH-TENSILE GALVANISED C-PURLINS (OTHER AVAILABLE SIZES)

The following purlin sizes provide an easy reference.

Section Identification	Dimensions					Mass kg/m
	A	B	C	D	t	
	mm	mm	mm	mm	mm	
SC100-16	100	50	50	14	1.6	2.78
SC100-20	100	50	50	14	2.0	3.45
SC150-16	150	65	65	16	1.6	3.84
SC150-20	150	65	65	16	2.0	4.77
SC150-25	150	65	65	16	2.5	5.93
SC200-16	200	75	75	20	1.6	4.82
SC200-20	200	75	75	20	2.0	6.00
SC200-25	200	75	75	20	2.5	7.46



DESIGN PRINCIPLE FOR PURLIN SELECTION TABLES

The purlin selection tables are derived basing on BS 5950 : Part 5 : 1987 "Code of Practice for design of cold formed sections".

- Purlin is simply supported with pinned joints over supports.
- Roof Pitch
The purlin selection tables are only applicable for roof pitch less than or equal to 30°.
- Loading
Besides dead load arising from roof self weight, the purlins are designed based on a live load of 0.25KN/m² (Malaysian Building Bylaw Clause 63) and wind load (uplift) of 0.50KN/m² respectively.
 - 1.4 D.L. + 1.6 L.L. (The most severe inward loading combination)
 - 1.0 D.L. + 1.4 W.L. (The most severe outward loading combination)
- Deflection
Under serviceability limit state the limiting deflection under total load is L/150 whereas under live load only is L/180 (where L is the purlin span).
- Design Strength P_y
The design strength P_y for purlins is to be taken as the minimum yield strength of 450N/mm²

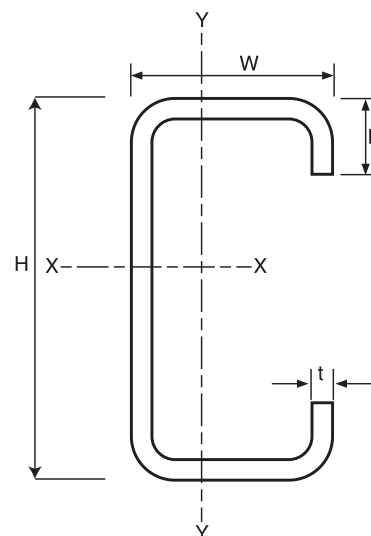


TABLE NO. 1 - PURLIN DIMENSIONS AND PROPERTIES

Section Identification	Dimensions				Mass per unit kg/m	Area mm ²	Effective section modulus		Full section modulus		Second moment of Area		Radius of Gyration	
	H	W	L	t			Zey	Zxx	Zyy	Ixx	Iyy	rxx	ryy	
	mm	mm	mm	mm			10 ³ mm ³	10 ³ mm ³	10 ³ mm ³	10 ³ mm ³	10 ⁶ mm ⁴	10 ⁶ mm ⁴	mm	mm
SC100-16	100	50	14	1.6	2.78	355	11.36	3.60	11.60	3.82	0.579	0.125	40.4	18.8
SC100-20	100	50	14	2.0	3.45	440	14.22	4.52	14.30	4.65	0.713	0.153	40.3	18.6
SC150-16	150	65	16	1.6	3.84	489	21.78	4.51	23.60	6.31	1.770	0.282	60.2	24.0
SC150-20	150	65	16	2.0	4.77	608	28.44	5.67	29.20	7.74	2.187	0.346	60.0	23.9
SC150-25	150	65	16	2.5	5.93	755	35.80	7.08	35.90	9.44	2.696	0.422	59.8	23.6
SC200-16	200	75	20	1.6	4.82	614	34.11	6.28	38.50	8.94	3.848	0.472	79.2	27.7
SC200-20	200	75	20	2.0	6.00	764	44.89	7.92	47.70	11.00	4.768	0.580	79.0	27.6
SC200-25	200	75	20	2.5	7.46	950	58.07	9.93	58.90	13.50	5.893	0.710	78.8	27.3

Note : Effective section modulus are calculated based on the effective cross section which taken into consideration of local buckling effect (Section Four BS5950 Part 5 : 1987).